



Tarun Kumar

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M.Tech - Data Science

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology, Guwahati	9.37	2023-25
PGDIIT	Indian Institute of Technology, Bombay	9.27	2022-23
B.Tech	Aryabhata Knowledge University	8.62	2018-22

EXPERIENCE

- DHL Supply Chain** Jul'25 – Present
Data Scientist Remote
 - Optimizing end-to-end supply chain operations using **machine learning, deep learning, and mathematical optimization** to enhance planning, inventory control, and logistics efficiency.
 - Designing and deploying **predictive and optimization models**, while collaborating cross-functionally to translate business challenges into **scalable, data-driven solutions** with measurable impact.
- ARKRAY, Japan** Feb'25 - July'25
Data Science Intern Remote
 - Developed a novel **class-aware attention mechanism** as a plug-and-play module for object detection models, achieving a **12% AP improvement** for minority classes and enhancing overall **mAP by 5%**.
 - Improved real-time detection efficiency by reducing inference time by **10%**, and boosted minority class performance using advanced data augmentation techniques such as **StyleGAN** and **ESRGAN**.

PROJECTS

- Optimizing Downstream NLP Tasks through Data Augmentation with LLMs** June'24 - June'25
Master's Thesis: Dr. Gaurav Trivedi(Dept. of EEE) and Dr.Teena Sharma(MFSDSAI), IIT Guwahati Report
 - Designed advanced data augmentation techniques to enhance model performance in **downstream NLP tasks** by fine-tuning language models on datasets like **SST-2** and **EMOTION**.
 - Achieved a **5%** improvement in overall accuracy on the SST dataset by implementing **Conditional Augmentation** using **RoBERTa** and **Llama-3.3-70B**, surpassing baseline performance.
- Care Vision AI: Disease Detection on Medical Imaging Datasets** Jan'24 - May'24
Course Project, IIT Guwahati GitHub
 - Built AI models for detecting medical conditions like skin cancer, brain tumors, lung cancer, and pneumonia using **CNN, SVM, Random Forest**, and **morphological operations**.
 - Enhanced **Recall** by **4%** using **Transfer Learning** with the **ResNet50** architecture and designed a user-friendly web application that seamlessly integrates these models, allowing efficient real-time detection and diagnosis.
- Large Language Model Based Healthcare Assistant** Jan'24 - Apr'24
Course Project, IIT Guwahati GitHub
 - Fine-tuned **Llama2-7b** model on a custom dataset from the **Medical Encyclopedia** to develop a **helthcare chatbot**.
 - Employed **QLoRA**, 4-bit quantization, and **Hugging Face** libraries for parameter-efficient fine-tuning, resulting in **50%** lower RAM usage and **400%** faster training times, significantly optimizing resource utilization.
- Steel Industry Energy Consumption Classification** Aug'23 - Nov'23
Course Project: Dr. Prithwijit Guha, Dept. of EEE, IIT Guwahati GitHub
 - Performed **Exploratory Data Analysis** on electricity consumption data from a steel plant with **35,000+** entries, and built machine learning models using **Logistic Regression, Random Forest, ANN, SVM**, and **Naive Bayes**.
 - Optimized the **Random Forest** model's performance through **hyperparameter tuning**, achieving an **F1 score of 0.98**.

TECHNICAL SKILLS

- Programming:** Python, C++, MATLAB*, R * Elementary proficiency
- Tools/Frameworks:** MySQL, Keras, NumPy, Pandas, TensorFlow, Pytorch, YOLO, LangChain, Docker

KEY COURSES TAKEN

- Mathematics:** Probability and Statistics, Matrix Computation, Optimization Techniques, Linear Algebra
- Data Science:** Deep Learning, Machine Learning, NLP with Large Language Models, Data Structures and Algorithms, DBMS, OOP, Data Visualization, Image Processing with ML

ACHIEVEMENTS

- Department Rank 1**, M.Tech Data Science, IIT Guwahati July'23 - July'25
- AIR 255**, Graduate Aptitude Test in Engineering (GATE) April'22